

Chapter Title: Hunger's chronicities: Covid and nutrition in a Mam-Maya community in Guatemala

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Book Title: Covid's Chronicities

Book Subtitle: From urgency to stasis in a pandemic era

Book Editor(s): Lenore Manderson, Nancy J. Burke

Published by: UCL Press. (2025)

Stable URL: <https://www.jstor.org/stable/jj.19724089.14>

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Hunger's chronicities: Covid and nutrition in a Mam-Maya community in Guatemala

Emily Yates-Doerr, Rosario García Mez
and María García Maldonado

For as long as anyone can remember, women in San Juan Ostuncalco have prepared a food for travelers. The base of *xwoiky*, as it was called in the Mam language spoken in the region, was masa – the soft dough made from boiled and ground maize. Women would roll this into large, flat rectangles which they cooked over a low-burning fire. Every so often they would flip the rectangles, slowly toasting them until they were dry and crisp – but never burnt. The toasted square would be left in the sun to dry even further and then, when there was no moisture left, the women would grind it, resulting in a food that Mercedes referred to with a smile as ‘the Corn Flakes of Guatemala’.¹ Men who trekked into the mountain for days at a time in search of firewood would take packages of *xwoiky* with them, as would those who left for the hard and painful labor at coastal plantations – a trip that would take several days as they descended 2,500 feet from their cold mountain villages to the hot, humid jungles near the sea. When it was time to break to eat, they would dissolve the *xwoiky* in coffee or hot or cold water, letting the dried maize expand. It would serve as the primary food for people as they traveled and it would last for weeks without spoiling.

Emily, imagining herself living on flakes of corn during days of hiking through the mountains, asked Mercedes if people didn't get hungry with only this to eat. Mercedes responded with an emphatic shake of her head. ‘If they only had bread to eat, then they would be hungry’. But *xwoiky*, made from their maize, was ‘rich’ and ‘delicious’ (see [Figure 8.1](#)).



Figure 8.1 Maize grown in San Juan Ostuncalco in 2022. Source: © Emily Yates-Doerr.

According to UNICEF, half of all children born in Guatemala suffer from chronic malnutrition (Escobar 2020). The World Bank backs up this claim, reporting that ‘Guatemala’s under five child malnutrition rate (at 47 percent) is among the ten highest worldwide’ (World Bank 2023). In Indigenous communities the reported rates are even higher, with government data suggesting that upwards of 70 percent of children in several mostly-Indigenous municipalities are chronically malnourished (USAID 2018). International organizations that monitor hunger note that cases are likely underreported. Data have been difficult to collect during the COVID-19 pandemic, but by many accounts, the pandemic has threatened Guatemalan food security, as supply chain failures combine with the effects of global warming to send food prices – and hunger rates – soaring. Maria Claudia Santizo, a nutrition specialist for UNICEF, emphasized that ‘child malnutrition is Guatemala’s greatest problem’, and explained the situation to journalist Lucía Escobar as follows:

When the coronavirus reached Guatemala, it had a strong impact on food security because of existing difficulties securing basic income: a lack of both formal and informal employment opportunities and

a downturn in remittances received by families from abroad. The result is less food in homes, leading to a decrease in available food for children. (Santizo, quoted in Escobar 2020)

There is a common association of hunger with thinness or even emaciation – consider, for example, the bleached hair and distended stomachs of children that are associated with the illness kwashiorkor (a form of severe protein malnutrition). But when it comes to chronic malnutrition, that is, malnutrition that accumulates over time, wasting – as measured through weight-for-age – is a notoriously capricious and unreliable calculation. Instead, global health experts assess chronic malnutrition through measurements of stature. They diagnose chronic malnutrition when a child (or population of children) is two standard deviations below the median global growth recommendations for height for age – a designation known as ‘stunting’. The concept of stunting, popularized by the British physiologist John Waterlow (1972), is premised on the idea that in conditions of nutrient deprivation, the bodies of growing children will conserve resources by becoming less tall. The rationale for equating stunting with hunger is that small stature reflects long-term and persistent undernutrition: low-grade, cumulative and irreversible damage that happens when children are not growing as they should. The World Health Organization (WHO 2015) notes that the consequences of stunting include poor cognition and educational performance, low adult wages, lost economic productivity and an increased risk of nutrition-related chronic disease later in life. Mercedes de Onis and Francisco Branca of the WHO describe childhood stunting as ‘the best overall indicator of children’s wellbeing and an accurate reflection of social inequalities’ (De Onis and Branca 2016, 12).

Our research on chronic malnutrition in San Juan questions the tendency to equate food insecurity, chronic malnutrition and stunting, drawing attention to shortcomings in existing global health frameworks for hunger. The field of public health nutrition and Mam speakers both distinguish between the felt condition of hunger and chronic malnutrition. Yet whereas nutritionists link chronic malnutrition to nutrient deficiency in maize-based Maya diet, people in San Juan emphasized how maize is filling and rich, and enough. As we illustrate, people did not report hunger during the pandemic, although they did have extensive stories of political and economic hardship. In many ways this resonates with scientists’ growing sense that what is frequently called chronic malnutrition is actually more a problem of stress – not of nutrition. Yet our research also questions this association. As we explain, Mam

speakers did not themselves focus on ‘stress’ but on social conditions, encompassing labor and land. We argue that as scientists work to evaluate hunger and malnutrition they must incorporate Indigenous knowledge into the science of global health.

The research setting

Our fieldwork took place near San Juan Ostuncalco, a region in Guatemala’s western highlands that reports some of the highest rates of chronic malnutrition in both the country and the world (Figure 8.2). The city center had a population of approximately 21,000 people at the 2018 census,² with twice as many people living in the two dozen rural communities that spread westward through rolling hillsides and steep ravines.

Nearly everyone identifies as Mam-Maya and speaks Mam as their first language. The origin story of the region, as relayed to us by a community member, was that an eagle living high in the mountains would capture the community’s children when their parents were absent, taking them to its nest where it would eat them. The ancestors of the



Figure 8.2 Women reach for tamales that they have prepared in anticipation of our visit. Source: © Emily Yates-Doerr

people of San Juan sought safety for their children by fleeing toward a lake, which they then blew away from the land with a giant straw to attempt to create a place to live.

That it took a degree of magic to make their homeland habitable resonates today. Rocky soils and intense commercial and governmental pressure to adopt Green Revolution technologies have led people to rely on agrochemicals to produce their food: maize, potatoes, black and broad beans and vegetables such as cabbage, radishes, onions, broccoli and cauliflower, much of which are exported internationally (Fischer and Benson 2006). The only health center is in the city center; social services are few and far between. Roads to the surrounding communities are in disrepair – washed out from increasingly heavy rains. Many houses lack plumbing, and children attend schools in barren under-resourced classrooms. While Mam is the predominant language of the region, many of the educators only speak Spanish, and many children stop attending school after just a few years. Mothers frequently discuss among themselves how teachers would hit or otherwise abuse their children.

Throughout colonial history, indentured servitude and anti-vagrancy laws forced almost all the men in the region to labor for mestizo or Spanish coastal plantations for much of the year. Today, the region continues to be pulled apart by migration, and stories of murdered children remain all too common. To overcome a pervasive sense of instability, roughly a quarter of the community – once only the men, but now also women and children – has left for the United States. Weathered adobe houses are interspersed with palatial cinder block homes, some that display the US flag, flying high in the breeze. We spoke with one woman with three children whose husband emigrated from Guatemala in 2019. She said she didn't know when he would be back, telling us 'he has to stay in the US because there is more he needs to do in Guatemala'. Her phrasing reflects the broad sense that international migration was the only way to sustain life at home.

We carried out our research in the summer of 2023, the first summer after national pandemic precautions were lifted.³ Several national and international maternal health organizations that had delivered food aid to pregnant and nursing women over the previous decade had ceased operations around the onset of Covid. Journalists had reported severe food restrictions and heightened food insecurity during the pandemic. A Reuters newspaper article authored by Guatemala-based journalists in May 2020 recounted that hunger was spreading under government restrictions, with Guatemalans flying white flags from their windows or

putting up desperate signs that signaled that they were in need of food assistance (Menchu and Renteria 2020).

We initially imagined the lockdown period of the pandemic in San Juan akin to the famous ‘Dutch Famine Winter’ during the Second World War, when people in the northern provinces of the Netherlands starved for months on end (Ramirez and Haas 2022). We thought that the quotidian, widespread food insecurity documented by development organizations would have been amplified during the pandemic, offering a meaningful opportunity to discuss the experience of living with chronic malnutrition. However, upon beginning our interviews, we quickly learned that we would have to adjust our expectations. The people we spoke with did not report feeling hunger.

The research question

The background question that animated our research was *What is chronic malnutrition for the people living through it?* This question might initially seem to be straightforward, but it is not. According to the science of stunting, the estimated 8.5 million people who are chronically malnourished in Guatemala may not, in fact, perceive this condition. Shortness, yes. But they may not correlate their height with the illnesses and deficiencies that scientists have linked to shortness, and they may never experience this shortness as *hunger*. In the English language, hunger is largely understood as a phenomenological condition: it is something felt in an individual’s body. But in global health discussions of chronic malnutrition people are ‘suffering’ (WHO 2024) from the consequences of hunger without knowing they are suffering. A UNICEF representative described chronic malnutrition as ‘the global crisis that you’ve never heard of’ (Lake 2012). The *chronicity* of chronic malnutrition normalizes it, making it imperceptible. This, in turn, offers a challenge for anthropological methods since they tend to rely on lived experience. We could not very easily ask people how they experience a condition that is not – or, at least, not necessarily – felt.

The movement between scientific and lived knowledge systems pertaining to chronic malnutrition is only part of the dilemma we faced. We also had to move between colonial languages (Spanish and English) and Mam-Maya, where direct translations for chronic malnutrition, stunting or food insecurity do not exist. That being said, it was the case that the concepts we were interested in were alive and active in San Juan in the form of development projects that focused on increasing children’s

height and nutrient supplementation programs to address food security. Women in San Juan are first-language Mam speakers, but they were familiar with programs to address *desnutrición crónica* and *inseguridad alimentaria* – the Spanish-language terms for chronic malnutrition and food insecurity (the technical Spanish-language terms for stunting, *demedro* or *retraso en el crecimiento*, were never used).

Still, we did not necessarily want to study Mam-Maya women's opinions of these programs. Instead, we wanted insight into how they might conceptualize and theorize the situation in Guatemala that scientists and public health officials were describing as chronic malnutrition. In their article *Words to Think With*, John Law and Annemarie Mol (2020) criticize the assumption that English-language concepts operate universally. They argue, 'When "international" academic conversations restrict themselves to using the intellectual tools available in English, they become limited and parochial' (264) and ask, noting the conceptual richness that can be afforded by working inter-linguistically, 'Might there not be further interesting words and ways of wording lying dormant in other languages?' (264). It further struck us as elitist – and possibly an inaccurate understanding of human culture – to think that people do not perceive or theorize the situations that shape their lives. If chronic malnutrition was a condition of suffering, we anticipated that people would have rich, nuanced ways of discussing it – although these theories might unfold in terms we didn't know and ways we didn't understand (see also Manderson and Smith Morris 2010).

The initial plan for our research was to talk with people about how they fed themselves and their families during the pandemic: What was a typical day of eating? Where did they procure and how did they prepare their food? What did they do when food ran out? What were their strategies of survival? We hoped that these prompts would lead to discussions of experiences of malnutrition or hunger that we could, together, analyze and discuss. We quickly learned that people did not report feeling hungry, but because our questions were open-ended this was not a problem. We could easily shift to engage in a discussion about food during the pandemic that would allow us to talk about aspects of malnutrition other than hunger.

Co-laboratory methods

In earlier research together we had developed a co-laboratory method for talking with women in San Juan about 'sustainable development' – a

term that poses many of the same problems as chronic malnutrition in the sense that it has come from outside the San Juan communities and is not commonly used locally (García Maldonado et al. 2016). Drawing from the work of Marisol de la Cadena (2015), the point of working together was not to reach a shared understanding. As she explains, co-laboring maintains attention on divergences in perspectives that exist as people interact. Rather than presume harmonious collaboration, co-laboring entails the labor of staying attentive to the equivocations that happen in translation across various systems of knowledge.

Emily Yates-Doerr is a white cultural and feminist anthropologist with settler-colonial ancestry and academic training in the United States and the Netherlands. Rosario García Meza is a mestiza Guatemalan nutrition scientist who holds a masters' degree in social justice research from Sussex University, UK and who frequently consults for international nutrition research in Guatemala. And María García Maldonado is a Mam-Maya lawyer who works as a cultural and linguistic translator for health and development projects in the region. Both Rosario and María maintain close ties to San Juan and Emily has carried out fieldwork in San Juan over the past decade. To learn about *sustainable development* we had previously spent time talking with women from San Juan about their lives in a way that did not use the words 'sustainable development' directly – for example, asking about their goals for the present and for the future and how they sought to maintain their communities over time. Recognizing their expertise in matters of intercultural exchange, we then explicitly talked about the translation of the term (Yates-Doerr 2025: 175–199).

For our question, *What is chronic malnutrition for the people living through it?* we took a similar approach of remaining curious about linguistic and conceptual frictions in the terms that people had to talk about eating, hunger and food. De la Cadena (2015) cautions that the method of co-laboring is not practiced to reach an understanding of the other's point of view, but instead to gain insight into all that is not known (see also Allen cited in Liboiron 2021, 22). We co-labor so that we maintain an awareness of the frictions, gaps and tensions between the concepts used in different communities of knowledge and practice.

The concept of *alimentarse* offers a good example. In Spanish, one word for nourishment is *alimentarse*, literally 'to give one's self *alimentos*'. Nourishment in English generally has a pleasant ring – nourishment is positive, caring or kind. Yet anthropologists working in rural Guatemala in the second half of the twentieth century, when scientists were popularizing the term *alimentarse*, encourage us to approach the term with caution. They note that the taxonomy of a food as '*alimento*' was a result

of the transfer of ‘information from Western medicine into the indigenous system’ (Cosminsky, in Diener 1982, 254). Anthropologist Paul Diener describes how the *Institute of Nutrition of Central America and Panama* worked hard to characterize foods high in protein – chicken, milk, eggs and its own manufactured for-profit maize-based nutrient supplements – as ‘muy [very] alimento’. As people began to classify all kinds of for-profit store-bought foods as ‘alimento’, including white flour and sugar, Diener came to see the concept as far more oppressive than emancipatory (1982, 254). He explained that ‘alimento’ imposed a colonialist, nutrient-based ethos over food, encouraging people to think of eating in terms of vitamin and mineral content – not in terms of the impacts on community relations, for example. Teachers, health educators and radio advertisements encouraged people to eat healthful ‘alimentos’. Meanwhile, he found that when Indigenous people categorized a food as *alimento* they were noting that its purchase required their labor to be ‘siphoned off by the regional market system into Ladino [non-Indigenous] channels’ (Collier, in Diener 1982, 258).

Diener’s analysis of the varied usages of the *alimento* concept helps to highlight the challenges and possibilities of methodologically tracing how the concepts and terminologies that surround food and nutrition circulate and are used in practice. Anthropologist Julián López García (2003) has likewise argued that Ch’orti’ Maya concepts of food and hunger diverge from the way these terms are conceptualized in the west. The 684-page Mam-Spanish dictionary of the Academia De Lenguas Mayas (2010) offers roughly seven uses of hunger: *Ch’ul* – the sound that hungry intestines make; *Jamtz’il* – to eat hastily because of hunger, such as a coyote devouring a sheep; *Jomjin* – an emptiness, such as that felt in the stomach when it is hungry; *Jul* – the suckling of babies when they are hungry; *Oyb’e’tz* – a snack eaten to stave off hunger; *Q’aq’il* – the experience of not having food and wanting to eat; and *Q’ol Wab’j* – the person whose job is to give food. Also included are expressions for food being sufficient, filling or absent.

María García explained that she regularly used two expressions for hunger when speaking in Mam. One referred to the harvest and climatic shifts. Her father, a farmer, would look at the coming rains or drought and speak of a time of hunger that might be coming to their community as crops failed. She used the expression *uti txilil wuaj* – literally, ‘there is no food’, and said it related broadly to an absence (in Spanish, *la falta de*). This focus on what was happening in the surroundings differed from the term *wiq’ij* – the phenomenological felt condition. *Wiq’ij* or *wiq’aq* transliterated as *hambre* in Spanish or ‘hunger’ in English. María

described *wiq'ij* or *wiq'aq* as the anxiety that 'we need something in the body' (*ansiedad que necesitamos algo en el cuerpo*). When we asked further about this anxiety, she explained that the phrase *nching'aq'ena* (*tengo hambre* or 'I am hungry') transliterates to *estoy quemandome* or 'I am burning inside'. She explained that *q'aq* means fuego or fire – so that the expression of being hungry might be thought of as having a fire in the belly.

The moment we switched registers to speak of hunger in terms of illness or a sickness she, like others with whom we spoke, switched to the Spanish-language term *desnutrición*, or in English, malnutrition. The fire-in-the-belly kind of hunger is related to a desire to eat; *desnutrición* occurred in the absence of this desire. In *desnutrición* the body, which was sick, did not feel hunger. Or, in slightly different terms, hunger is the feeling that happens when you do not have tortillas to eat whereas *desnutrición* is a sickness that results when you stop wanting them. Maya people lacking maize to eat would be hungry; Maya people who did not want maize but filled themselves on bread would become *desnutridos* (malnourished).

To explore these conceptual frictions we carried out open-ended interviews with seven women and one man who was a university professor in the nearby city of Quetzaltenango. All people we interviewed were Mam-Maya and lived in the communities surrounding San Juan Ostuncalco. At first, we kept the questions we posed broad: How did you maintain yourself and your family during the pandemic? Tell us about what you ate and how were your meals? Following the co-laboratory methods described above, when people used terms that we found to be relevant in either Spanish or Mam, we often stopped to talk with them more about these terms. For example, when a woman said her maize was filling we asked her to explain the sensation (María and she both translated the sensation of a full-belly feel to Spanish as *llenarse* which seemed fairly equivalent to the English-language expression of 'being full'). When a woman spoke of not feeling afraid of becoming hungry during the pandemic (in the phenomenological sense of *wiq'ij*) we asked more about this and she switched to an environmental register to discuss hunger, explaining she did not fear hunger because she had land, and with this, maize. Our conversations frequently moved between Spanish and Mam, with María leading discussions about how to best translate Mam terms with the people with whom we spoke – all of whom could converse in Spanish, though they preferred to speak in Mam.

To complement these conversations, we also held a cooking-based focus group for women to discuss their experiences of cooking and

feeding their families during the pandemic as we prepared a meal together. We coordinated this gathering with a community leader, who had worked with the several NGOs in the region that had previously aimed to improve chronic malnutrition among pregnant women and new mothers. For our research on the local semantics of chronic malnutrition we did not need to restrict recruitment to new mothers. As a result, the community leader drew upon her contacts to assemble a group that also included grandmothers. In the end, thirteen women (and roughly twice as many children) from four of the surrounding communities arrived at her house to prepare a lunchtime meal of chicken and noodle soup flavored with epazote (a regional green, leafy herb). All women in attendance had children, except one who was pregnant for the first time. The women ranged in age from 19 to 55 years old. Reflecting the general pattern in the community, 11 of the 13 women had immediate family members living in the United States. Every woman brought tamales (made by wrapping maize dough tightly in banana leaves to be boiled), which were added to a community pot. We prepared the soup together, using the occasion to talk about past cooking experiences, with a focus on the early months of the pandemic.

We also attended one women's health meeting – part of a project run by an NGO working in San Juan to build solidarity and reduce stress among Mam women designated as vulnerable and in need of extra support. A community midwife we interviewed held the meeting at her home, and invited women to join the conversations with us and for us to also participate in her event. The meetings were a chance for women to process their life experiences with each other. When we attended, the midwife helped us to ask the women to reflect on their experiences feeding themselves and their families through the pandemic.

'The time of sickness'

On 13 March 2020, Guatemala closed. For several months, only medical workers, ambulance drivers and some supermarket food distributors were allowed to leave their homes. People in urban areas, with the means to do so, ordered food in regulated quantities from online distributors. Others followed the mandated hours that they were allowed to access grocery stores, which utilized a quota system to control entry. Schools closed for two years, as did offices. Some retail stores that did not traditionally distribute food began to do so as an excuse to open, but for many months during what people today call *el tiempo de la enfermedad* ('the

time of sickness'), many people in urban centers of the country were afraid and stayed inside.⁴

In rural communities, the effects of the pandemic were felt in other ways. People feared contagion, and did not want to break the law: at 4 p.m., when the national curfew went into effect, they generally returned to their homes. But they did not isolate indoors as did people who lived in the cities. In San Juan nearly everyone grows maize, the foundation of all their meals and of their community life. The major annual harvest is in November, so when the pandemic closed the country they still had ample supplies. In February 2020, as people around the world began to eye the emerging story of Covid, people in San Juan were busy preparing and tilling the earth. In March, when the world shut down, they were waiting for the rains to signal that it was time to begin to plant maize for the next cycle of harvest. When we asked if they stayed at home when the country went into lockdown, they said this would be impossible. They were afraid of sickness, but foregoing a cycle of planting would mean certain death.

Similarly, women tended not to isolate themselves from their aging mothers. When we asked at the cooking group whether women took the precaution of staying away from elderly family members, the women unanimously shook their heads. They knew this was the guidance but they relied too much on intergenerational ties to do so. Public health mandates to isolate from social networks were not designed with poor communities like theirs in mind. These were guidelines for middle class or wealthy families – not for people, like those living in the communities around San Juan, who needed to farm and needed each other to live.

Conventional economic understandings of hunger would have us write, 'despite their poverty not a single person we spoke to reported hunger'. Yet it was also *because of* their poverty that they could not adhere to the mandates of lockdown and continued to farm. The people we spoke with knew that COVID was dangerous and that it had contributed to deaths in the community, but they saw no option: they had to work together to facilitate the production and distribution of food. One mother had nine children to feed and no land of her own. Her husband had previously commuted long distances for his work in construction; public transportation closures meant he lost his job. But in her case, as with others in her position, neighbors with land stepped in to provide food. Another woman with six children who similarly did not own land reported that she switched from buying local maize to buying maize from the coast, which was cheaper. When this ran out, she was cared for by community members, who shared their maize with her. Food was

scarce, but as with every other person we spoke with, this woman said that during the pandemic she and her children never found themselves without food. The pandemic was a time of fear and sadness, but the maize people in San Juan had stored in 2019 would get them through 2020, and the maize they grew in 2020 would get them through 2021. By all reports, the pandemic was not a time of hunger.

In fact, several people mentioned that lockdown restrictions caused them to spend more time farming on their land than they had in previous years. Many San Juan households grow their maize in small plots disconnected from their homes. In addition to the time they spent at these plots, several women also started or tended to home gardens, filling these with broad beans, peas, and the many common green, leafy herbs that people grew and ate (for example, cilantro, peppermint, epazote, parsley, huauzontle). Children still ate chips and lollipops, much as they had before the pandemic.⁵ And eventually, the government stepped in to distribute food aid for families with school-age children that included processed and packaged foods such as oil, cornmeal, pasta, sugar-fortified cereal, rice, and beans. But these boxes arrived infrequently and food prices escalating rapidly. Access to city grocery stores was made difficult by the lack of transportation, and people in San Juan relied on home-grown foods more so than usual.

While they did not run out of maize, women did mention struggling with the closing of the public *molinas*. Typically they pay a small fee to the owners of these machines to mill corn into the masa that was the basis of the region's staple foods served at every meal. Without access to the mills, they had to roll corn out by hand, which was laborious and time-consuming work. Although this did not result in hunger, it added to the challenges of daily life during the pandemic. Our interviews were filled with stories of pandemic-related hardship: these included husbands losing their jobs because they could no longer travel by public transportation to work; electricity being shut off because they could not pay their bills; family members who had migrated to the United States just before the pandemic who stopped replying to calls for several months; family members who became sick, children who became sick; and many community members who died from suspected-COVID-19 and from other illnesses – exactly what those might be they could only guess, since they did not have professional medical help.⁶

In the background of these descriptions of pandemic marginalization is historic and widespread discrimination against Mam-Maya people. During Guatemala's armed conflict, the military targeted Maya people because of their cultural practices. The killing or disappearing of

more than 200,000 people included an estimated 83 per cent who were Maya (CEH 1998). The war officially ended with the signing of the Peace Accords in 1996. But the acceleration of Indigenous migration since the late 1990s suggests that ethnic and racist violence did not end. Today, Maya people continue to face widespread discrimination in accessing government services and employment. During Covid, by drawing upon their plots of land and their ties to their community they were able to feed themselves; hunger was not their problem. Still, they suffered in many other ways.

During the cooking group, Marta mentioned that her eldest three teenage children – two daughters and a son – had left for the United States during the pandemic. She said this casually and without apparent sadness, then adding that her 15-year-old son, who had joined us for lunch that day, would probably join his older siblings in Florida soon. Emily's two children were also at that moment in the United States, and she empathized with the difficulty of separation. Marta's neighbor, sitting next to us, added that after Marta's children had migrated, Marta had cried every hour of every day for months on end. The neighbor speculated that because it was common for children to migrate, this made it harder, not easier, on their mothers, who lived in fear of their children's imminent departure. Anthropologist Briana Nichols (2021), who has worked in a nearby community, has documented that many children, who witness the suffering and loneliness that results from emigration, want to stay. Nonetheless, the older they became, the more likely it was that they would leave.

Fear of their departure hangs over the process of children aging, and then once they leave, those who stay behind are left to worry about their children's disappearance or death. Just days before our visit, the body of a 23-year-old man was repatriated from the United States to San Juan for burial. Neighbors said that his death was an accident, but they had heard rumors he had suffered considerably from economic and ethnic discrimination in the preceding years. A few days later, another young person from San Juan – Mercedes Carolina Izara Vicente – died in Arlington, Virginia, shortly after becoming sick. At the funeral in the United States, which was broadcast to her Guatemalan community on Facebook, the pastor spoke of how hard it is for migrants to find appropriate health care.

But even though the trip to the United States is dangerous and life there for migrants is often extremely perilous, staying in San Juan was not a viable choice for many. The schoolteacher we interviewed told us about how much he worried about the community's youth.

Young people, he explained, had a terrible burden placed on them: they had few opportunities for educational advancement and the possibilities for local employment were demanding, low-wage jobs. These didn't pay enough for them to support their families or themselves. He spoke about the difficulty that young people in the community experienced as their life opportunities shifted north. He mentioned one young boy in the community who became lost to the helplessness of his future during the pandemic, committing suicide by drinking poison.

One mother we interviewed told us of leaving her two younger children behind in San Juan in 2021 so she could travel to the United States to reconnect with her older children who had emigrated before the pandemic began. She was apprehended by authorities before reaching them, and deported with significant debt back to Guatemala. By all accounts, it is money sent from abroad from family members – not government support – that keeps the communities of San Juan going. An article in *The Guardian* reports that remittances rose 35 per cent between 2020 and 2021 (Abbott 2022). The US\$15 billion-plus dollars sent to Guatemala from abroad in 2021 played an integral role in sustaining Guatemalan families during the pandemic. As journalist Jeff Abbott wrote, rural Guatemalans 'depend on remittances for survival while elites benefit from low labour costs' (2022).

The US government ended asylum proceedings at the start of the pandemic, citing a public health policy passed in 1944 that allowed federal authorities to prohibit migration so as to not spread disease. For 38 months, the United States turned away all Guatemalans who might have claimed asylum – and yet, still people in San Juan went north. Everyone knew that those traveling to the United States often experienced tremendous hardship. People were not leaving because they wanted to go, but because they found it impossible to stay. The pandemic had not left them hungry. Their maize was strong; they knew how to feed themselves. Instead, they were leaving their homelands because the pandemic had exacerbated already difficult social conditions including economic discrimination and violence against Maya people. People generally did not want to leave, but they saw no other choice.

Conclusion: a measure of societies

Chronic malnutrition is an indicator that measures a situation, not of children, but of societies.

Lesley Ramírez (Quoted in Redacción Con Criterio 2020)

Western scientists have long held that Indigenous Guatemalan diets are deficient in nutrients. For years, nutritionists were convinced that a maize-based diet lacked protein, causing people to live with moderate malnutrition that, while imperceptible, nonetheless wreaked havoc on their fitness, their intelligence and, ultimately, their lives. For the better part of the second half of the twentieth century, major attempts were made to fortify Guatemalan diets with protein, until many scientists began to realize that protein deficiency had never been a widespread problem.

Some degree of micronutrient deficiency may have existed in the ancestral Guatemalan diet (iodizing salt has helped to prevent goiter; as has adding vitamin A to sugar helped to reduce cases of blindness; see Ruz and Solomons 2019; Arroyave et al. 1981). But it is also the case that the corn-bean-squash-chile based diet, the staple food of Indigenous communities, is remarkably healthy in terms of nutrient composition, containing the nutrients people need not just to live but to thrive.

This was an observation made in the early 1950s by Guatemalan anthropologist Antonio Goubaud Carrerra, who argued that nutritionists' disdain for the Maya diet was driven by their Western ideology. His research on the food practices of Indigenous Maya people led to a 'public revelation' that the typical diet of rural Indigenous Guatemalans was not actually 'as bad as recently thought' (Adams 2016, 81). In her retrospective account of Goubaud Carrerra's work, anthropologist Abigail Adams writes that Goubaud Carrerra criticized those who held racist explanations for why Indigenous people were shorter than Europeans. Refuting the idea that the Maya diet was inferior to the Western diet, Goubaud Carrerra published a newspaper article in which he argued,

Do not the Holy Scriptures say, 'Man does not live by bread alone?' If we find that the material bread of indigenous people is fine, then it would be the height of ignorance to think that their spirit must receive whole wheat bread. (Goubaud Carrerra cited in Adams 2016, 81)

After we concluded our research in San Juan, Emily spoke with Dr Noel Solomons, who runs a nutrition research center in Guatemala (where Rosario García Meza also works). Studies conducted at his center have pointed to the nutritional merits of the traditional Guatemalan diet (Valdés-Ramos et al. 2001) and he characterized the idea that Guatemala is a country with endemic food insecurity as 'misinformation'. While global warming could certainly radically upend food supplies, he argued that the research coming from Guatemala does not point to *chronic malnutrition* – understood as persistent food insecurity – as a historic problem in the

country. Rather, he suggested that what is imprecisely called chronic malnutrition is not caused by a deficiency in available nutrients but rather by an environmental trigger that he referred to as ‘stress’ (see also Solomons 2019). He explained, ‘the problem that women and children experience is that of early life “stress exposures” that can compromise physiological development’. The problem that policymakers are calling *chronic malnutrition* does not actually relate to nutrient intake, and the problem ‘will only go away when these environmental stressors are overcome’.

He pointed us toward recent research using large datasets of child growth to analyze the relation between stunting and wasting. This research, affiliated with the Bill and Melinda Gates Foundation, the Emergency Nutrition Network and the Knowledge Integration Child Growth Consortium (Mertens et al. 2023; Benjamin-Chung et al. 2023; Sadler et al. 2022), suggests that unless a child experiences a stress event – such as an infection or acute wasting – being short, also known as stunting, is a benign experience. In other words, shortness on its own is not a problem and it should not be equated to chronic malnutrition.

In important ways, this ‘cutting edge’ scientific analysis is an echo of what we heard from community members in San Juan: the problems they struggled with during the pandemic were not principally those of food scarcity or hunger but, rather, were those of extreme and persistent sociopolitical hardship. Consider, for example, the filling, delicious meal of *xwoky*. For centuries, Mam women have prepared this for men in their communities ahead of their travel to plantations. Recall Mercedes’ insistence that *xwoky* was a tasty food that could sustain people over long distances of traveling and periods of time. Her reframing of *xwoky* as delicious – not deficient – helps us to see that the problem scientists were calling chronic malnutrition was not a problem of Maya physiology or Maya foods but a problem of colonial dispossession and the brutality of labor that Indigenous people were forced, by law, to undertake. The science of chronic malnutrition has tended to focus on inadequacy of bodies and diets. In naming *xwoky* as a food of abundance, Mercedes was helping us to recenter our stories of malnutrition on the harms of a political and economic system that saw Maya bodies as expendable and did not care if Maya people died.

Geographer Julie Guthman (2013) writes of how nutrition scientists and policymakers frequently contribute to what she describes as ‘problem closure’ in their frameworks for malnutrition. She explains: ‘a specific definition of a problem and socially acceptable solutions are used to frame the study of the problem’s causes and consequences’ (2013,142). In the case of Guatemalan malnutrition, global health workers regularly

embed taken-for-granted assumptions about the deficiency of the Maya diet into their models of health, foreclosing other explanations for why people suffer.

We argued above that the refusal of women of San Juan to categorize *xwoky* as insufficient is instructive; it helps us to see how malnutrition's origins lie not in maize but in the conditions of labor. As scientists begin to shift their focus away from 'nutrients in foods' to 'stress in bodies', we might also take heed of the disinclination of Mam speakers to use the term *éstres* (stress) – a term that has no Maya language equivalent. Women certainly know and can use the Spanish language term for stress, but we find its absence telling: scientific discussions of 'stress' run the risk of biologizing hardship, turning problems of social injustice into problems of bodies and hormones.

When describing the hardships they encountered in the pandemic, people we interviewed in San Juan kept their focus squarely on conditions of labor and land. They spoke of the pandemic as a time of challenge and difficulty, but the *chronicity* of chronic malnutrition lay in the persistence of political structures that disenfranchized Maya people – not in the persistence of bodily hunger. People did not run out of maize during the pandemic. When some families struggled with provisions, other families stepped in. The concerns they relayed to us were those of enduring exploitation, and the related challenge of building community when so many families and neighbors were separated, spread thousands of miles apart. For centuries, state laws forced San Juan men to labor on coastal plantations where they produced the sugar for wealthy landowners (Oglesby 2013). Eventually the government would fortify the sugar with micronutrients that scientists said Indigenous bodies lacked, and sugar companies would sell the sugar they helped to produce back to them with marketing that described the product as 'healthy'. Today, many of the community's residents – including their children – are still forced to migrate to work, as a living wage remains out of reach in San Juan.

Uti txilil wuaj – or the climatic condition of being without food – is devastating and deadly, but *nchinq'aq'ena*, 'fire in the belly', is not necessarily bad. Just as managed fire is crucial to the regenerative milpa agriculture used to grow maize for millenia, *nchinq'aq'ena* is a feeling that comes in anticipation of a feast. In contrast, not feeling hunger – *desnutrición* – is a clear condition of illness. This happens when medical care is impossible to access, when children have untreated parasites, when infection endures, when communities fall apart or when there is nothing to eat but candies or bread. The chronicity of malnutrition from which people suffer was not to be found in their bodily tissues, but in

political structures that devalue their lives. In the words of Guatemalan nutritionist Lesley Ramirez, ‘Chronic malnutrition is an indicator that measures a situation, not of children, but of societies’ (2020).

As scientists debate the equation between stunting, stress and food security they might take heed of San Juan women’s conceptualization of chronic malnutrition as a sociopolitical problem – an indicator of societies. The shift in this framing also shifts the terrain of possibility for solutions. Instead of asking how to fortify foods with nutrients or how to make people taller, they might instead ask how to repair the harms, not only of the pandemic but of history. How do we ensure that people have land on which to grow their food? How do we redress economic discrimination and racist violence? When the pandemic is finally over, how can we support people to gather with their families and communities to feast? (Figure 8.3). People in San Juan know how to care for hunger – they do so through their land and community. But today, much of their community has traveled so far away that they no longer have access to *xwo’ky* to extinguish the fire in their bellies. Without community there can be no maize and without maize there can be no community. Without both they will find themselves in a time of absence – a season of hunger, *la falta de, uti txilil wuaj*.



Figure 8.3 Women in San Juan Ostuncalco gather around a meal of chicken noodle soup in 2023. Source: © Emily Yates-Doerr.

Notes

- 1 We use pseudonyms for the people with whom we spoke in San Juan Ostuncalco. Ethical approval for this research was granted by the Ethics Review Board of the Amsterdam Institute of Social Science Research.
- 2 See Guatemala census data at <https://www.citypopulation.de/en/guatemala/cities/>. Accessed 28 September 2024.
- 3 Covid was still present but vaccines were widely available and there were no longer quarantine or curfew measures in place (masks had been required until 18 July 2022 and on public transportation until 17 October 2022 – see online updates at <https://bit.ly/3N5USIG>). Accessed 28 September 2024.
- 4 A mask mandate was in place until the summer of 2022; masks were widely used inside and outside and not highly politicized as they were in the United States. Even after the mandate ended, many people continued to wear masks.
- 5 At our cooking group, shortly before soup was served, an ice cream truck pulled through the neighborhood and the cooking activities came to a halt as women and children purchased cones to eat. Without judgment – the three of us will also enjoy an ice cream cone on a hot day – we asked about whether kids ate more or less candies and chips during the pandemic than before it. The unanimous reply was that the consumption of chips and candy remained the same.
- 6 The subject of food arose when discussing pandemic-related deaths. Normally ritual foods are served at funerals, but during the pandemic it was impossible to gather to eat together and collectively mourn. The concern, however, was that without collective eating the healing process would be interrupted, and not that maize had run out leaving people hungry.

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